

JAMP ITRACONAZOLE ORAL SOLUTION 10MG/ML

Itraconazole Oral Solution 10mg/mL

DESCRIPTION

Yellow transparent solution.

COMPOSITION

1 mL JAMP Itraconazole Oral Solution contains 10 mg itraconazole.

PHARMACODYNAMICS

Mechanism of action

In vitro studies have demonstrated that itraconazole impairs the synthesis of ergosterol in fungal cells. Ergosterol is a vital cell membrane component in fungi. Impairment of its synthesis ultimately results in an antifungal effect.

Pharmacokinetic (PK)/Pharmacodynamic (PD) relationship

The PK/PD relationship for itraconazole, and for triazoles in general, is poorly understood.

Pharmacodynamic effects

Microbiology

Itraconazole, a triazole derivative, has a broad spectrum of activity.

For itraconazole, interpretive breakpoints have not been established by CLSI for the *Candida* spp. and filamentous fungi.

EUCAST breakpoints for itraconazole have been established for *Aspergillus flavus*, *A. fumigatus*, *A. nidulans* and *A. terreus*, and are as follows: susceptible ≤ 1 mg/L, resistant > 1 mg/L. EUCAST breakpoints for itraconazole have been established for *Candida albicans* and *C. dubliniensis*, and are as follows: susceptible ≤ 0.06 mg/L, resistant > 0.06 mg/L. EUCAST breakpoints for itraconazole have been established for *Candida parapsilosis* and *C. tropicalis*, and are as follows: susceptible ≤ 0.125 mg/L, resistant > 0.125 mg/L. Interpretive breakpoints have not been established by EUCAST for *Candida glabrata*, *C. krusei*, *C. guilliermondii*, *Cryptococcus neoformans*, and Non-species related breakpoints for *Candida*.

In vitro studies demonstrate that itraconazole inhibits the growth of a broad range of fungi pathogenic for humans at concentrations usually ≤ 1 μ g/ml. These include:

Aspergillus spp., *Blastomyces dermatitidis*, *Cladosporium* spp., *Coccidioides immitis*, *Cryptococcus neoformans*, *Geotrichum* spp., *Histoplasma* spp., including *H. capsulatum*, *Paracoccidioides brasiliensis*, *Penicillium marneffeii*, *Sporothrix schenckii* and *Trichosporon* spp. Itraconazole also displayed activity *in vitro* against *Epidermophyton floccosum*, *Fonsecaea* spp., *Malassezia* spp., *Microsporium* spp., *Pseudallescheria boydii*, *Trichophyton* spp. and various other yeasts and fungi.

The principal fungus types that are not inhibited by itraconazole are *Zygomycetes* (e.g., *Rhizopus* spp., *Rhizomucor* spp., *Mucor* spp. and *Absidia* spp.), *Fusarium* spp., *Scedosporium* spp. and *Scopulariopsis* spp.

Azole resistance appears to develop slowly and is often the result of several genetic mutations. Mechanisms that have been described are overexpression of ERG11, which encodes the target enzyme 14 α -demethylase, point mutations in ERG11 that lead to decreased target affinity and/or transporter overexpression resulting in increased efflux. Cross-resistance between members of the azole class has been observed within *Candida* spp., although resistance to one member of the class does not necessarily confer resistance to other azoles. Itraconazole-resistant strains of *Aspergillus fumigatus* have been reported.

PHARMACOKINETIC

General pharmacokinetic characteristics

Peak plasma concentrations are reached within 2.5 hours following administration of the oral solution. As a consequence of non-linear pharmacokinetics, itraconazole accumulates in plasma during multiple dosing. Steady-state concentrations are generally reached within about 15 days, with $C_{\max}$ and AUC values 4 to 7-fold higher than those seen after a single dose. Steady-state $C_{\max}$ values of about 2 $\mu\text{g/ml}$ are reached after oral administration of 200 mg once daily. The terminal half-life of itraconazole generally ranges from 16 to 28 hours after single dose and increases to 34 to 42 hours with repeated dosing. Once treatment is stopped, itraconazole plasma concentrations decrease to an almost undetectable concentration within 7 to 14 days, depending on the dose and duration of treatment. Itraconazole mean total plasma clearance following intravenous administration is 278 ml/min. Itraconazole clearance decreases at higher doses due to saturable hepatic metabolism.

Absorption

Itraconazole is rapidly absorbed after administration of the oral solution. Peak plasma concentrations of itraconazole are reached within 2.5 hours following administration of the oral solution under fasting conditions. The observed absolute bioavailability of itraconazole under fed conditions is about 55% and increases by 30% when the oral solution is taken in fasting conditions.

Itraconazole exposure is greater with the oral solution than with the capsule formulation when the same dose of drug is given. (See *Warnings and Precautions*.)

Distribution

Most of the itraconazole in plasma is bound to protein (99.8%) with albumin being the main binding component (99.6% for the hydroxy- metabolite). It has also a marked affinity for lipids. Only 0.2% of the itraconazole in plasma is present as free drug. Itraconazole is distributed in a large apparent volume in the body (> 700 L), suggesting extensive distribution into tissues. Concentrations in lung, kidney, liver, bone, stomach, spleen and muscle were found to be two to three times higher than corresponding concentrations in plasma, and the uptake into keratinous tissues, skin in particular, up to four times higher. Concentrations in the cerebrospinal fluid are

much lower than in plasma, but efficacy has been demonstrated against infections present in the cerebrospinal fluid.

Metabolism

Itraconazole is extensively metabolized by the liver into a large number of metabolites. *In vitro* studies have shown that CYP3A4 is the major enzyme involved in the metabolism of itraconazole. The main metabolite is hydroxy-itraconazole, which has *in vitro* antifungal activity comparable to itraconazole; though plasma concentrations of this metabolite are about twice those of itraconazole.

Excretion

Itraconazole is excreted mainly as inactive metabolites in urine (35%) and in feces (54%) within one week of an oral solution dose. Renal excretion of itraconazole and the active metabolite hydroxy-itraconazole account for less than 1% of an intravenous dose. Based on an oral radiolabeled dose, fecal excretion of unchanged drug ranges from 3% to 18% of the dose.

As re-distribution of itraconazole from keratinous tissues appears to be negligible, elimination of itraconazole from these tissues is related to epidermal regeneration. Contrary to plasma, the concentration in skin persists for 2 to 4 weeks after discontinuation of a 4-week treatment and in nail keratin – where itraconazole can be detected as early as 1 week after start of treatment – for at least six months after the end of a 3-month treatment period.

Special populations

Hepatic impairment

Itraconazole is predominantly metabolized in the liver. Data are not available in cirrhotic patients during long-term use of itraconazole. (See also *Dosage and Administration*, and *Warnings and Precautions*.)

Renal impairment

Limited data are available on the use of oral itraconazole in patients with renal impairment. Data are not available in renally impaired patients during long-term use of itraconazole. Dialysis has no effect on the half-life or clearance of itraconazole or hydroxy-itraconazole. (See also *Dosage and Administration*, and *Warnings and Precautions*.)

Pediatrics

Limited pharmacokinetic data are available on the use of itraconazole in the pediatric population.

Hydroxypropyl- β -Cyclodextrin

The oral bioavailability of hydroxypropyl- β -cyclodextrin given as a solubilizer of itraconazole in oral solution is on average lower than 0.5% and is similar to that of hydroxypropyl- β -cyclodextrin alone. This low oral bioavailability of hydroxypropyl- β -cyclodextrin is not modified by the presence of food and is similar after single and repeated administrations.

INDICATIONS

JAMP Itraconazole Oral Solution is indicated:

- For the treatment of oral and/or esophageal candidosis in HIV-positive or other immunocompromised patients.
- As prophylaxis of deep fungal infections when standard therapy is considered in appropriate in patients with hematological malignancy or undergoing bone marrow transplant, and who are expected to become neutropenic (i.e., < 500 cells/ μ L).

At present there are insufficient data on clinical efficacy in the prevention of aspergillosis.

CONTRAINDICATIONS

- JAMP Itraconazole Oral Solution is contraindicated in patients with known hypersensitivity to itraconazole or to any of the excipients.
- Coadministration of a number of CYP3A4 substrates is contraindicated with JAMP Itraconazole Oral Solution. Increased plasma concentrations of these drugs, caused by coadministration with itraconazole, may increase or prolong both therapeutic and adverse effects to such an extent that a potentially serious situation may occur. For example, increased plasma concentrations of some of these drugs can lead to QT prolongation and ventricular tachyarrhythmias including occurrences of torsade de pointes, a potentially fatal arrhythmia. Specific examples are listed in *Interactions*.
- JAMP Itraconazole Oral Solution should not be administered to patients with evidence of ventricular dysfunction such as congestive heart failure (CHF) or a history of CHF except for the treatment of life-threatening or other serious infections. (See *Warnings and Precautions*.)
- JAMP Itraconazole Oral Solution must not be used during pregnancy (except for life-threatening cases). (See *Pregnancy, Breast-feeding and Fertility*.)

Women of childbearing potential taking JAMP Itraconazole Oral Solution should use contraceptive precautions. Highly effective contraception should be continued until the menstrual period following the end of JAMP Itraconazole Oral Solution therapy.

WARNINGS AND PRECAUTIONS

Cardiac effects

Itraconazole should not be used in patients with congestive heart failure or with a history of congestive heart failure unless the benefit clearly outweighs the risk. This individual benefit/risk assessment should take into consideration factors such as the severity of the indication, the dosing regimen, (e.g., total daily dose), and individual risk factors for congestive heart failure. These risk factors include cardiac disease, such as ischemic and valvular disease; significant pulmonary disease, such as chronic obstructive pulmonary disease; and renal failure and other edematous disorders. Such patients should be informed of the signs and symptoms of congestive heart failure, should be treated with caution, and should be monitored for signs and symptoms of congestive heart failure during treatment; if such signs or symptoms do occur during treatment, Itraconazole should be discontinued.

Calcium channel blockers can have negative inotropic effects which may be additive to those of itraconazole. In addition itraconazole can inhibit the metabolism of calcium channel blockers. Therefore, caution should be used when co-administering itraconazole and calcium channel blockers due to an increased risk of CHF.

Interaction potential

Coadministration of specific drugs with itraconazole may result in changes in efficacy of itraconazole and/or the coadministered drug, life-threatening effects and/or sudden death. Drugs that are contraindicated, not recommended or recommended for use with caution in combination with itraconazole are listed in *Interactions*.

Cross-hypersensitivity

There is limited information regarding cross-hypersensitivity between itraconazole and other azole antifungal agents. Caution should be used in prescribing JAMP Itraconazole Oral Solution to patients with hypersensitivity to other azoles.

Neuropathy

If neuropathy occurs that may be attributable to JAMP Itraconazole Oral Solution, the treatment should be discontinued.

Hearing loss

Transient or permanent hearing loss has been reported in patients receiving treatment with itraconazole. Several of these reports included concurrent administration of quinidine which is contraindicated (see *Contraindications*). The hearing loss usually resolves when treatment is stopped, but can persist in some patients.

Cross-resistance

In systemic candidosis, if fluconazole-resistant strains of *Candida* species are suspected, it cannot be assumed that these are sensitive to itraconazole, hence it is recommended to have their sensitivity tested before the start of itraconazole therapy.

Interchangeability

It is not recommended that Itraconazole capsule and oral solution to be used interchangeably. This is because drug exposure is greater with the oral solution than with the capsules when the same dose of drug is given.

Hepatic effects

Very rare cases of serious hepatotoxicity, including some cases of fatal acute liver failure, have occurred with the use of Itraconazole. Most of these cases involved patients who, had preexisting liver disease, were treated for systemic indications, had significant other medical conditions and/or were taking other hepatotoxic drugs. Some patients had no obvious risk factors for liver disease. Some of these cases were observed within the first month of treatment, including some within the first week. Liver function monitoring should be considered in patients receiving Itraconazole treatment. Patients should be instructed to promptly report to their physician signs and symptoms

suggestive of hepatitis such as anorexia, nausea, vomiting, fatigue, abdominal pain or dark urine. In these patients treatment should be stopped immediately and liver function testing should be conducted.

Limited data are available on the use of oral itraconazole in patients with hepatic impairment. Caution should be exercised when the drug is administered in this patient population. It is recommended that patients with impaired hepatic function be carefully monitored when taking itraconazole.

In patients with elevated or abnormal liver enzymes or active liver disease, or who have experienced liver toxicity with other drugs, treatment with Itraconazole is strongly discouraged unless there is a serious or life threatening situation where the expected benefit exceeds the risk. It is recommended that liver function monitoring should be done in patients with pre-existing hepatic function abnormalities or those who have experienced liver toxicity with other medications. (See *Pharmacokinetic Properties - Special Populations, Hepatic impairment.*)

Cystic fibrosis

In cystic fibrosis patients, variability in therapeutic levels of itraconazole was observed. If a patient does not respond to Itraconazole oral solution, consideration should be given to switching to intravenous or to alternative therapy.

Pediatrics

Clinical data on the use of Itraconazole oral solution in pediatric patients are limited. The use of Itraconazole oral solution in pediatric patients is not recommended unless it is determined that the potential benefit outweighs the potential risks.

Elderly

Clinical data on the use of Itraconazole oral solution in elderly patients are limited. It is advised to use Itraconazole oral solution in these patients only if it is determined that the potential benefit outweighs the potential risks. In general, it is recommended that the dose selection for an elderly patient should be taken into consideration, reflecting the greater frequency of decreased hepatic, renal, or cardiac function, and of concomitant disease or other drug therapy.

Renal impairment

Limited data are available on the use of oral itraconazole in patients with renal impairment. The exposure of itraconazole may be lower in some patients with renal insufficiency. Caution should be exercised when this drug is administered in this patient population and adjusting the dose may be considered.

Treatment of severely neutropenic patients

Itraconazole oral solution as treatment for oral and/or esophageal candidosis was not investigated in severely neutropenic patients. Due to the pharmacokinetic properties (see *Pharmacokinetic Properties*), Itraconazole oral solution is not recommended for initiation of treatment in patients at immediate risk of systemic candidosis.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

No studies on the effects on the ability to drive and use machines have been performed. When driving vehicles and operating machinery the possibility of adverse reactions such as dizziness, visual disturbances and hearing loss (See *Adverse Reactions.*), which may occur in some instances, must be taken into account.

PREGNANCY, BREAST-FEEDING AND FERTILITY PREGNANCY

Itraconazole must not be used during pregnancy except for life-threatening cases where the potential benefit to the mother outweighs the potential harm to the fetus (see *Contraindications*).

Women of childbearing potential

Women of childbearing potential taking Itraconazole should use contraceptive precautions. Highly effective contraception should be continued until the menstrual period following the end of Itraconazole therapy.

Breast-feeding

A very small amount of itraconazole is excreted in human milk. The expected benefits of treatment with Itraconazole should therefore be weighed against the potential risk of breastfeeding. In case of doubt, the patient should not breast-feed.

Fertility

There is no evidence of a primary influence on fertility under treatment with itraconazole.

HP- β -CD has no antifertile effect.

INTERACTIONS

Itraconazole is a drug with a high interaction potential. The various types of interaction and associated general recommendations are described below. In addition, a table is provided listing examples of drugs that may interact with itraconazole, organized per drug family for easy reference. This list of examples is not comprehensive and therefore the label of each drug that is coadministered with itraconazole should be consulted for information related to the route of metabolism, interaction pathways, potential risks, and specific actions to be taken with regards to coadministration.

Itraconazole is mainly metabolized through CYP3A4. Other substances that either share this metabolic pathway or modify CYP3A4 activity may influence the pharmacokinetics of itraconazole. Coadministration of itraconazole with moderate or potent CYP3A4 inducers may decrease the bioavailability of itraconazole and hydroxy-itraconazole to such an extent that efficacy may be reduced. Coadministration with moderate or potent inhibitors of CYP3A4 may increase the bioavailability of itraconazole, which may result in increased or prolonged pharmacologic effects of itraconazole.

Itraconazole and its major metabolite, hydroxy-itraconazole are potent CYP3A4 inhibitors. Itraconazole is an inhibitor of the drug transporters P-glycoprotein and breast cancer resistance

protein (BCRP). Itraconazole can inhibit the metabolism of drugs metabolized by CYP3A4 and can inhibit the drug transport by P-glycoprotein and/or BCRP, which may result in increased plasma concentrations of these drugs and/or their active metabolite(s) when they are administered with itraconazole. These elevated plasma concentrations may increase or prolong both therapeutic and adverse effects of these drugs. For some drugs, coadministration with itraconazole may result in decreased plasma concentrations of the drug or of the active moiety of the drug. This may result in reduced efficacy of the drug.

Following cessation of medical treatment with itraconazole, plasma concentrations decrease below the detection limit within 7 to 14 days, depending on the dose and duration of treatment. In patients with hepatic cirrhosis or in subjects receiving CYP3A4 inhibitors the plasma concentrations decline slower. This is particularly important for consideration when initiating therapy with drugs whose metabolism is affected by itraconazole.

The following general recommendations apply, unless stated differently in table.

- ‘Contraindicated’: Under no circumstances is the drug to be coadministered with itraconazole. This applies to:
 - CYP3A4 substrates for which increased plasma concentrations may increase or prolong therapeutic and/or adverse effects to such an extent that a potentially serious situation may occur. (see *Contraindications*)
- ‘Not recommended’: It is recommended that the use of the drug be avoided, unless the benefits outweigh the potentially increased risks. If coadministration cannot be avoided, clinical monitoring is recommended, and the dosage of itraconazole and/or the coadministered drug adapted as deemed necessary. When appropriate, it is recommended that plasma concentrations be measured. This applies to:
 - Moderate or potent CYP3A4 inducers: not recommended from 2 weeks before and during treatment with itraconazole
 - CYP3A4/P-gp/BCRP substrates for which increased or decreased plasma concentrations result in significant risk: not recommended during and up to 2 weeks after treatment with itraconazole.
- ‘Use with caution’: Careful monitoring is recommended when the drug is coadministered with itraconazole. Upon coadministration, it is recommended that patients be monitored closely and the dosage of itraconazole and/or the coadministered drug adapted as deemed necessary. When appropriate, it is recommended that plasma concentrations be measured.

This applies to:

- Moderate or potent inhibitors of CYP3A4
- CYP3A4/P-gp/BCRP substrates for which increased or decreased plasma concentrations result in a clinically relevant risk

Examples of interacting drugs are listed in the table below. The drugs listed in this table are based on either drug interaction studies or case reports, or potential interactions based on the mechanism of interaction.

Medicinal products within class	Expected/Potential effect on drug levels (see footnotes for additional info)	Clinical comment (see codes above for additional info)
Alpha Blockers		
Alfuzosin Silodosin Tamsulosin	Alfuzosin C _{max} (↑↑), AUC (↑↑) ^a Silodosin C _{max} (↑↑), AUC (↑↑) ^a Tamsulosin C _{max} (↑↑), AUC (↑↑) ^a	Not recommended during and for 2 weeks after treatment with itraconazole. Increased risk of alfuzosin/silodosin/tamsulosin-related adverse reactions ^c .
Analgesics		
Alfentanil Buprenorphine (IV and sublingual) Oxycodone Sufentanil	Alfentanil AUC (↑↑ to ↑↑↑↑) ^a Buprenorphine C _{max} (↑↑), AUC (↑↑) ^a Oxycodone C _{max} ↑, AUC ↑↑ Sufentanil conc increase (extent unknown) ^{a,b}	Use with caution, monitor for adverse reactions related to the analgesic ^c , dose reduction of alfentanil/buprenorphine/oxycodone/sufentanil may be necessary.
Fentanyl	Fentanyl IV AUC (↑↑) ^a Fentanyl other form. conc increase (extent unknown) ^{a,b}	Not recommended during and for 2 weeks after treatment with itraconazole. Increased risk of fentanyl-related adverse reactions ^c .
Levacetylmethadol (levomethadyl)	Levacetylmethadol C _{max} (↑↑), AUC (↑↑↑) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of levacetylmethadol-related adverse reactions, such as QT prolongation and TdP.
Methadone	(R)-methadone C _{max} (↑), AUC (↑) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of methadone-related adverse reactions, such as potentially life-threatening respiratory depression, QT prolongation and TdP.
Antiarrhythmics		
Digoxin	Digoxin C _{max} ↑, AUC ↑	Use with caution, monitor for digoxin adverse reactions, dose reduction of digoxin may be necessary ^c .
Disopyramide	Disopyramide conc increase (↑↑) ^{a,b}	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of disopyramide-related adverse reactions, such as serious arrhythmias including TdP.

Medicinal products within class	Expected/Potential effect on drug levels (see footnotes for additional info)	Clinical comment (see codes above for additional info)
Dofetilide	Dofetilide C _{max} (↑), AUC (↑) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of dofetilide-related adverse reactions, such as serious ventricular arrhythmias including TdP.
Dronedarone	Dronedarone C _{max} (↑↑↑), AUC (↑↑↑↑) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of dronedarone-related adverse reactions, such as QT prolongation and cardiovascular death.
Quinidine	Quinidine C _{max} ↑, AUC ↑↑	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of quinidine-related adverse reactions, such as QT prolongation, TdP, hypotension, confusion and delirium.
Antibacterials		
Bedaquiline	Bedaquiline C _{max} (↔), AUC (↑) during 2 weeks of bedaquiline q.d. dosing ^a	Not recommended, coadministration for more than 2 weeks at any time during bedaquiline dosing is not recommended: increased risk of bedaquiline-related adverse reactions ^c .
Ciprofloxacin Erythromycin	Itraconazole C _{max} ↑, AUC ↑	Use with caution, monitor for itraconazole adverse reactions, dose reduction of itraconazole may be necessary.
Clarithromycin	Clarithromycin conc increase (extent unknown) ^{a,b} Itraconazole C _{max} ↑, AUC ↑;	Use with caution, monitor for adverse reactions related to itraconazole and/or clarithromycin ^c , dose reduction of itraconazole and/or clarithromycin may be necessary.
Delamanid Trimetrexate	Delamanid conc. increase (extent unknown) ^{a,b} Trimetrexate conc increase (extent unknown) ^{a,b}	Use with caution, monitor for delamanid/trimetrexate adverse reactions ^c , dose reduction of delamanid/trimetrexate may be necessary.
Isoniazid Rifampicin	Isoniazid: itraconazole conc. (↓↓↓) ^{a,b} Rifampicin: itraconazole AUC ↓↓↓	Not recommended from 2 weeks before and during treatment with itraconazole, Itraconazole efficacy may be reduced.

Medicinal products within class	Expected/Potential effect on drug levels (see footnotes for additional info)	Clinical comment (see codes above for additional info)
Rifabutin	Rifabutin conc. increase (extent unknown) ^{a,b} Itraconazole: C _{max} ↓↓, AUC ↓↓	Not recommended from 2 weeks before, during and for 2 weeks after treatment with itraconazole. Itraconazole efficacy may be reduced and increased risk of rifabutin-related adverse reactions ^c .
Telithromycin	In healthy subjects: telithromycin C _{max} ↑, AUC ↑ In severe renal impairment: telithromycin AUC (↑↑) ^a In severe hepatic impairment: telithromycin conc. increase (extent unknown) ^{a,b}	Contraindicated in patients with severe renal or hepatic impairment during and for 2 weeks after treatment with itraconazole. Increased risk of telithromycin-related adverse reactions, such as hepatotoxicity, QT prolongation and TdPsc. Use with caution in other patients, monitor for telithromycin adverse reactions, dose reduction of telithromycin may be necessary.
Anticoagulants and Antiplatelet Drugs		
Apixaban Edoxaban Rivaroxaban Vorapaxar	Apixaban C _{max} (↑), AUC (↑) ^a Edoxaban C _{max} (↑), AUC (↑) ^a Rivaroxaban C _{max} (↑), AUC (↑ to ↑↑) ^a Vorapaxar C _{max} (↑), AUC (↑) ^a	Not recommended during and for 2 weeks after treatment with itraconazole. Increased risk of apixaban/edoxaban/rivaroxaban/vorapaxar-related adverse reactions ^c .
Coumarins (e.g., warfarin) Cilostazol	Coumarins (eg, warfarin) conc increase (extent unknown) ^{a,b} Cilostazol C _{max} (↑), AUC (↑↑) ^a	Use with caution, monitor for coumarins/cilostazol adverse reactions ^c , dose reduction of coumarins/cilostazol may be necessary.
Dabigatran	Dabigatran C _{max} (↑↑), AUC (↑↑) ^a	Use with caution, monitor for dabigatran adverse reactions ^c , dose reduction of dabigatran may be necessary.
Ticagrelor	Ticagrelor C _{max} (↑↑), AUC (↑↑↑) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of ticagrelor-related adverse reactions, such as bleeding.
Anticonvulsants		

Medicinal products within class	Expected/Potential effect on drug levels (see footnotes for additional info)	Clinical comment (see codes above for additional info)
Carbamazepine	Carbamazepine conc. ($\uparrow$) ^{a,b} Itraconazole conc. ($\downarrow\downarrow$) ^{a,b}	Not recommended from 2 weeks before, during and for 2 weeks after treatment with itraconazole. Itraconazole efficacy may be reduced and increased risk for carbamazepine-related adverse reactions ^c .
Phenobarbital Phenytoin	Phenobarbital: itraconazole conc. ($\downarrow\downarrow\downarrow$) ^{a,b} Phenytoin: itraconazole AUC $\downarrow\downarrow\downarrow$	Not recommended from 2 weeks before and during treatment with itraconazole. Itraconazole efficacy may be reduced.
Antidiabetics		
Repaglinide Saxagliptin	Repaglinide $C_{\max}$ $\uparrow$, AUC $\uparrow$ Saxagliptin $C_{\max}$ ($\uparrow\uparrow$), AUC ($\uparrow\uparrow$) ^a	Use with caution, monitor for repaglinide/saxagliptin adverse reactions ^c , dose reduction of repaglinide/saxagliptin may be necessary.
Antihelmintics, antifungals and antiprotozoals		
Artemether-lumefantrine Quinine	Artemether $C_{\max}$ ($\uparrow\uparrow$), AUC ($\uparrow\uparrow$) ^a Lumefantrine $C_{\max}$ ($\uparrow$), AUC ($\uparrow$) ^a Quinine $C_{\max}$ $\leftrightarrow$, AUC $\uparrow$	Use with caution, monitor for artemether-lumefantrine/quinine adverse reactions ^c . Refer to the label for specific actions to be taken.
Halofantrine	Halofantrine conc increase (extent unknown) ^{a,b}	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of halofantrine-related adverse reactions, such as QT prolongation and fatal arrhythmias.
Isavuconazole	Isavuconazole $C_{\max}$ ($\leftrightarrow$), AUC ($\uparrow\uparrow\uparrow$) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of isavuconazole-related adverse reactions, such as hepatic adverse reactions, hypersensitivity reactions and embryo-fetal toxicity.
Praziquantel	Praziquantel $C_{\max}$ ($\uparrow\uparrow$), AUC ($\uparrow$) ^a	Use with caution, monitor for praziquantel adverse reactions ^c , dose reduction of praziquantel may be necessary.
Antihistamines		
Astemizole	Astemizole $C_{\max}$ ($\uparrow$), AUC ($\uparrow\uparrow$) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of astemizole-related adverse reactions, such as QT prolongation, TdP and other ventricular arrhythmias.

Medicinal products within class	Expected/Potential effect on drug levels (see footnotes for additional info)	Clinical comment (see codes above for additional info)
Bilastine Ebastine Rupatadine	Bilastine C _{max} (↑↑), AUC (↑) ^a Ebastine C _{max} ↑↑, AUC ↑↑↑ Rupatadine conc increase (↑↑↑↑) ^{a,b}	Use with caution, monitor for bilastine/ebastine/rupatadine adverse reactions ^c , dose reduction of bilastine/ebastine/rupatadine may be necessary.
Mizolastine	Mizolastine C _{max} (↑), AUC (↑) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of mizolastine-related adverse reactions, such as QT prolongation.
Terfenadine	Terfenadine conc increase (extent unknown) ^b	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of terfenadine-related adverse reactions, such as QT prolongation, TdP and other ventricular arrhythmias.
Antimigraine Drugs		
Eletriptan	Eletriptan C _{max} (↑↑), AUC (↑↑↑) ^a	Use with caution, monitor for eletriptan adverse reactions ^c , dose reduction of eletriptan may be necessary.
Ergot alkaloids (such as dihydroergotamine, ergometrine, ergotamine, methylergometrine)	Ergot alkaloids conc increase (extent unknown) ^{a,b}	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of ergot alkaloid-related adverse reactions, such as ergotism.
Antineoplastics		
Bortezomib Brentuximab vedotin Busulfan Erlotinib Gefitinib Imatinib Ixabepilone Nintedanib Panobinostat Ponatinib Ruxolitinib	Bortezomib AUC (↑) ^a Brentuximab vedotin AUC (↑) ^a Busulfan C _{max} ↑, AUC ↑ Erlotinib C _{max} (↑↑), AUC (↑) ^a Gefitinib C _{max} ↑, AUC ↑	Use with caution, monitor for adverse reactions related to the antineoplastic drug ^c , dose reduction of the antineoplastic drug may be necessary.

Medicinal products within class	Expected/Potential effect on drug levels (see footnotes for additional info)	Clinical comment (see codes above for additional info)
Sonidegib Tretinoin (oral) Vandetanib	Imatinib C _{max} (↑), AUC (↑) ^a Ixabepilone C _{max} (↔), AUC (↑) ^a Nintedanib C _{max} (↑), AUC (↑) ^a Panobinostat C _{max} (↑), AUC (↑) ^a Ponatinib C _{max} (↑), AUC (↑) ^a Ruxolitinib C _{max} (↑), AUC (↑) ^a Sonidegib C _{max} (↑), AUC (↑↑) ^a Tretinoin C _{max} (↑), AUC (↑) ^a Vandetanib C _{max} ↔, AUC ↑	
Idelalisib	Idelalisib C _{max} (↑), AUC (↑) ^a Itraconazole serum conc. increase (extent unknown) ^{a,b}	Use with caution, monitor for adverse reactions related to itraconazole and/or idelalisib ^c , dose reduction of itraconazole and/or idelalisib may be necessary.
Axitinib Bosutinib Cabazitaxel Cabozantinib Ceritinib Cobimetinib Crizotinib Dabrafenib Dasatinib Docetaxel Glasdegib Ibrutinib Lapatinib Nilotinib Olaparib Pazopanib Sunitinib Talazoparib	Axitinib C _{max} (↑), AUC (↑↑) ^a Bosutinib C _{max} (↑↑↑), AUC (↑↑↑) ^a Cabazitaxel C _{max} (↔), AUC (↔) ^a Cabozantinib C _{max} (↔), AUC (↑) ^a Ceritinib C _{max} (↑), AUC (↑↑) ^a Cobimetinib C _{max} ↑↑, AUC ↑↑↑ Crizotinib C _{max} (↑), AUC (↑↑) ^a Dabrafenib AUC (↑) ^a Dasatinib C _{max} (↑↑), AUC (↑↑) ^a	Not recommended during and for 2 weeks after treatment with itraconazole. Increased risk of adverse reactions related to the antineoplastic drug ^c . Additionally: For cabazitaxel, even though the change in pharmacokinetic parameters did not reach statistical significance in a low-dose drug interaction study with ketoconazole, a high variability in the results was observed. For ibrutinib, refer to the label for specific actions to be taken.

Medicinal products within class	Expected/Potential effect on drug levels (see footnotes for additional info)	Clinical comment (see codes above for additional info)
Trabectedin Trastuzumab emtansine Vinca alkaloids	Docetaxel AUC ($\leftrightarrow$ to $\uparrow\uparrow$) ^a Glasdegib C _{max} ($\uparrow$), AUC ($\uparrow\uparrow$) ^a Ibrutinib C _{max} ($\uparrow\uparrow\uparrow$), AUC ($\uparrow\uparrow\uparrow$) ^a Lapatinib C _{max} ($\uparrow\uparrow$), AUC ($\uparrow\uparrow$) ^a Nilotinib C _{max} ($\uparrow$), AUC ($\uparrow\uparrow$) ^a Olaparib C _{max} $\uparrow$, AUC $\uparrow\uparrow$ Pazopanib C _{max} ($\uparrow$), AUC ($\uparrow$) ^a Sunitinib C _{max} ($\uparrow$), AUC ($\uparrow$) ^a Talazoparib C _{max} $\uparrow$, AUC $\uparrow$ ^a Trabectedin C _{max} ($\uparrow$), AUC ($\uparrow$) ^a Trastuzumab emtansine conc increase (extent unknown) ^{a,b} Vinca alkaloid conc increase (extent unknown) ^{a,b}	
Regorafenib	Regorafenib AUC ($\downarrow\downarrow$ by estimation of active moiety) ^a	Not recommended during and for 2 weeks after treatment with itraconazole. Regorafenib efficacy may be reduced.
Irinotecan	Irinotecan and its active metabolite conc increase (extent unknown) ^{a,b}	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of irinotecan-related adverse reactions, such as potentially life-threatening myelosuppression and diarrhea.
Venetoclax	Venetoclax C _{max} ($\uparrow\uparrow\uparrow$), AUC ($\uparrow\uparrow\uparrow$) ^a	Contraindicated for chronic lymphocytic leukemia/small lymphocytic lymphoma patients during dose initiation/titration/ramp-up phase of venetoclax. Otherwise, not

Medicinal products within class	Expected/Potential effect on drug levels (see footnotes for additional info)	Clinical comment (see codes above for additional info)
		recommended during and for 2 weeks after treatment with itraconazole.
Antipsychotics, Anxiolytics and Hypnotics		
Alprazolam Aripiprazole Brotizolam Buspirone Cariprazine Haloperidol Midazolam (iv) Perospirone Quetiapine Ramelteon Risperidone Suvorexant Zopiclone	Alprazolam C_{max} ↔, AUC ↑↑ Aripiprazole C_{max} ↑, AUC ↑ Brotizolam C_{max} ↔, AUC ↑↑ Buspirone C_{max} ↑↑↑↑, AUC ↑↑↑↑ Cariprazine (↑↑) ^{a,b} Haloperidol C_{max} ↑, AUC ↑ Midazolam (iv) conc increase ↑↑ ^b Perospirone C_{max} ↑↑↑, AUC ↑↑↑ Quetiapine C_{max} (↑↑), AUC (↑↑↑) ^a Ramelteon C_{max} (↑), AUC (↑) ^a Risperidone conc increase ↑ ^b Suvorexant C_{max} (↑), AUC (↑↑) ^a Zopiclone C_{max} ↑, AUC ↑	Use with caution, monitor for adverse reactions related to the antipsychotic, anxiolytic or hypnotic drug ^c , dose reduction of these drugs may be necessary.
Lurasidone	Lurasidone C_{max} (↑↑↑), AUC (↑↑↑) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of lurasidone-related adverse reactions, such as hypotension, circulatory collapse, severe extrapyramidal symptoms, seizures.
Midazolam (oral)	Midazolam (oral) C_{max} ↑ to ↑↑, AUC ↑↑ to ↑↑↑↑	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of midazolam-related adverse reactions, such as respiratory depression, cardiac arrest, prolonged sedation and coma.
Pimozide	Pimozide C_{max} (↑), AUC (↑↑) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of pimozide-related adverse reactions, such as

Medicinal products within class	Expected/Potential effect on drug levels (see footnotes for additional info)	Clinical comment (see codes above for additional info)
		cardiac arrhythmias, possibly associated with QT prolongation and TdP.
Sertindole	Sertindole conc increase (extent unknown) ^{a,b}	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of sertindole-related adverse reactions, such as QT prolongation and TdP.
Triazolam	Triazolam C _{max} ↑ to ↑↑, AUC ↑↑ to ↑↑↑↑	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of triazolam-related adverse reactions, such as seizures, respiratory depression, angioedema, apnea and coma.
Antivirals		
Asunaprevir (boosted) Tenofovir disoproxil fumarate (TDF)	Asunaprevir C _{max} (↑↑↑), AUC (↑↑↑) ^a Tenofovir conc increase (extent unknown) ^{a,b}	Use with caution, however, refer to the label of the antiviral drug for specific actions to be taken.
Boceprevir	Boceprevir C _{max} (↑), AUC (↑↑) ^a Itraconazole conc increase (extent unknown) ^{a,b}	Use with caution, monitor for adverse reactions related to itraconazole and/or boceprevir ^c , dose reduction of itraconazole may be necessary. Refer to the boceprevir label for specific actions to be taken.
Cobicistat	Cobicistat conc increase (extent unknown) ^{a,b} Itraconazole conc increase (extent unknown) ^{a,b}	Use with caution, monitor for adverse reactions related to itraconazole, dose reduction of itraconazole may be necessary.
Daclatasvir Vaniprevir	Daclatasvir C _{max} (↑), AUC (↑↑) ^a Vaniprevir C _{max} (↑↑↑), AUC (↑↑↑) ^a	Use with caution, monitor for daclatasvir/vaniprevir adverse reactions ^c , dose reduction of daclatasvir/vaniprevir may be necessary.
Darunavir (boosted) Fosamprenavir (ritonavir-boosted) Telaprevir	Ritonavir-boosted darunavir: itraconazole C _{max} (↑↑), AUC (↑↑) ^a Ritonavir-boosted fosamprenavir: itraconazole C _{max} (↑), AUC (↑↑) ^a	Use with caution, monitor for itraconazole adverse reactions, dose reduction of itraconazole may be necessary.

Medicinal products within class	Expected/Potential effect on drug levels (see footnotes for additional info)	Clinical comment (see codes above for additional info)
	Telaprevir: itraconazole $C_{\max}$ (↑), AUC (↑↑) ^a	
Elvitegravir (boosted)	Elvitegravir $C_{\max}$ (↑), AUC (↑) ^a Itraconazole conc increase (extent unknown) ^{a,b}	Use with caution, monitor for adverse reactions related to itraconazole and/or elvitegravir (ritonavir-boosted) ^c . Dose reduction of itraconazole may be necessary; refer to the elvitegravir label for specific actions to be taken.
Efavirenz Nevirapine	Efavirenz: itraconazole $C_{\max}$ ↓, AUC ↓ Nevirapine: itraconazole $C_{\max}$ ↓, AUC ↓↓	Not recommended from 2 weeks before and during treatment with itraconazole. Itraconazole efficacy may be reduced.
Elbasvir/Grazoprevir	Elbasvir $C_{\max}$ (↔), AUC (↑) ^a Grazoprevir $C_{\max}$ (↔), AUC (↑↑) ^a	Use with caution, monitor for adverse reactions related to the co-administered drugs ^c . Refer to the elbasvir/grazoprevir label for specific actions to be taken.
Glecaprevir/Pibrentasvir	Glecaprevir $C_{\max}$ (↑↑), AUC (↑↑ to ↑↑↑) ^a Pibrentasvir $C_{\max}$ (↔ to ↑), AUC (↔ to ↑↑) ^a	Use with caution, monitor for adverse reactions related to the co-administered drugs ^c . Refer to the glecaprevir/pibrentasvir label for specific actions to be taken.
Indinavir	Itraconazole conc. ↑ ^b Indinavir $C_{\max}$ ↔, AUC ↑	Use with caution, monitor for adverse reactions related to itraconazole and/or indinavir ^c , dose reduction of itraconazole and/or indinavir may be necessary.
Maraviroc	Maraviroc $C_{\max}$ (↑↑), AUC (↑↑↑) ^a	Use with caution monitor for adverse reactions ^c . Dose reduction of maraviroc may be necessary.
Ombitasvir/Paritaprevir/ Ritonavir with or without Dasabuvir	Itraconazole $C_{\max}$ (↑), AUC (↑↑) ^a Ombitasvir $C_{\max}$ (↔), AUC (↑) ^a Paritaprevir $C_{\max}$ (↑), AUC (↑↑) ^a	Use with caution, monitor for adverse reactions related to itraconazole and/or the antivirals ^c , dose reduction of itraconazole may be necessary. Refer to the label(s) of the coadministered drugs for specific actions to be taken.

Medicinal products within class	Expected/Potential effect on drug levels (see footnotes for additional info)	Clinical comment (see codes above for additional info)
	Ritonavir $C_{\max}$ (↑), AUC (↑) ^a Dasabuvir $C_{\max}$ (↑), AUC (↑) ^a	
Ritonavir	Itraconazole $C_{\max}$ (↑), AUC (↑↑) ^a Ritonavir $C_{\max}$ (↔), AUC (↑) ^a	Use with caution, monitor for adverse reactions related to itraconazole and/or ritonavir ^c . Dose reduction of itraconazole may be necessary; refer to the ritonavir label for specific actions to be taken.
Saquinavir	Saquinavir (unboosted) $C_{\max}$ ↑↑, AUC ↑↑↑ Itraconazole (with boosted saquinavir) $C_{\max}$ (↑), AUC (↑↑) ^a	Use with caution, monitor for adverse reactions related to itraconazole and/or saquinavir ^c . Dose reduction of itraconazole may be necessary; refer to the saquinavir label for specific actions to be taken.
Beta Blockers		
Nadolol	Nadolol $C_{\max}$ ↑↑, AUC ↑↑	Use with caution, monitor for nadolol adverse reactions ^c . Dose reduction of nadolol may be necessary.
Calcium Channel Blockers		
Bepridil	Bepridil conc increase (extent unknown) ^{a,b}	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of bepridil-related adverse reactions, such as new arrhythmias and TdP type ventricular tachycardia.
Diltiazem	Diltiazem & Itraconazole conc increase (extent unknown) ^{a,b}	Use with caution, monitor for adverse reactions related to itraconazole and/or diltiazem ^c , dose reduction of itraconazole and/or diltiazem may be necessary.
Felodipine Lercanidipine Nisoldipine	Felodipine $C_{\max}$ ↑↑↑, AUC ↑↑↑ Lercanidipine AUC (↑↑↑↑) ^a Nisoldipine $C_{\max}$ (↑↑↑↑), AUC (↑↑↑↑) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of dihydropyridine-related adverse reactions, such as hypotension and peripheral edema.
Other dihydropyridines Verapamil	Dihydropyridine conc increase (extent unknown) ^{a,b}	Use with caution, monitor for dihydropyridine/verapamil adverse reactions ^c , dose reduction of dihydropyridine/verapamil may be necessary.

Medicinal products within class	Expected/Potential effect on drug levels (see footnotes for additional info)	Clinical comment (see codes above for additional info)
	Verapamil conc increase (extent unknown) ^{a,b}	
Cardiovascular Drugs, Misc		
Aliskiren Riociguat Sildenafil (pulmonary hypertension) Tadalafil (pulmonary hypertension)	Aliskiren C _{max} ↑↑↑, AUC ↑↑↑ Riociguat C _{max} (↑), AUC (↑↑) ^a Sildenafil/Tadalafil conc increase (extent unknown but effect may be greater than reported under Urological Drugs) ^{a,b}	Not recommended during and for 2 weeks after treatment with itraconazole ^c . Increased risk of adverse reactions related to the cardiovascular drug.
Bosentan Guanfacine	Bosentan C _{max} (↑↑), AUC (↑↑) ^a Guanfacine C _{max} (↑), AUC (↑↑) ^a	Use with caution, monitor for bosentan/guanfacine adverse reactions ^c , dose reduction of bosentan/guanfacine may be necessary.
Ivabradine	Ivabradine C _{max} (↑↑), AUC (↑↑↑) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of ivabradine-related adverse reactions, such as atrial fibrillation, bradycardia, sinus arrest and heart block.
Ranolazine	Ranolazine C _{max} (↑↑), AUC (↑↑) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of ranolazine-related adverse reactions, such as QT prolongation and renal failure.
Contraceptives*		
Dienogest Ulipristal	Dienogest C _{max} (↑), AUC (↑↑) ^a Ulipristal C _{max} (↑↑), AUC (↑↑↑) ^a	Use with caution, monitor for contraceptive adverse reactions ^c , refer to the dienogest/ulipristal label for specific actions to be taken.
Diuretics		
Eplerenone	Eplerenone C _{max} (↑), AUC (↑↑↑) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of eplerenone-related adverse reactions, such as hyperkalemia and hypotension.

Medicinal products within class	Expected/Potential effect on drug levels (see footnotes for additional info)	Clinical comment (see codes above for additional info)
Finerenone	Finerenone C _{max} (↑↑), AUC (↑↑↑) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of finerenone-related adverse reactions ^c .
Gastrointestinal Drugs		
Aprepitant Loperamide Netupitant	Aprepitant AUC (↑↑↑) ^a Loperamide C _{max} ↑↑, AUC ↑↑ Netupitant C _{max} (↑), AUC (↑↑) ^a	Use with caution, monitor for aprepitant/loperamide/netupitant adverse reactions ^c . Dose reduction of aprepitant/loperamide may be necessary. Refer to the netupitant label for specific actions to be taken.
Cisapride	Cisapride conc increase (extent unknown) ^{a,b}	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of cisapride-related adverse reactions, such as serious cardiovascular events including QT prolongation, serious ventricular arrhythmias and TdP.
Domperidone	Domperidone C _{max} ↑↑, AUC ↑↑	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of domperidone-related adverse reactions, such as serious ventricular arrhythmias and sudden cardiac death.
Naloxegol	Naloxegol C _{max} (↑↑↑), AUC (↑↑↑↑) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of naloxegol-related adverse reactions, such as opioid withdrawal symptoms.
<i>Saccharomyces boulardii</i>	<i>S. boulardii</i> colonization decrease (extent unknown)	Not recommended during and for 2 weeks after treatment with itraconazole. <i>S. boulardii</i> efficacy may be reduced.

Voclosporin	Voclosporin C _{max} (↑↑↑), AUC (↑↑↑↑) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of voclosporin-related adverse reactions ^c .
Lipid Regulating Drugs		
Atorvastatin	Atorvastatin C _{max} ↔ to ↑↑, AUC ↑ to ↑↑	Use with caution, monitor for atorvastatin adverse reactions ^c . Dose reduction of atorvastatin may be necessary.
Lomitapide	Lomitapide C _{max} (↑↑↑↑), AUC (↑↑↑↑) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of lomitapide-related adverse reactions, such as hepatotoxicity and severe gastrointestinal reactions.
Lovastatin Simvastatin	Lovastatin C _{max} ↑↑↑↑, AUC ↑↑↑↑ Simvastatin C _{max} ↑↑↑↑, AUC ↑↑↑↑	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of lovastatin/ simvastatin-related adverse reactions, such as myopathy, rhabdomyolysis and liver enzyme abnormalities.
Nonsteroidal Anti-Inflammatory Drugs		
Meloxicam	Meloxicam C _{max} ↓↓, AUC ↓	Use with caution, monitor for reduced efficacy of meloxicam, dose adaption of meloxicam may be necessary.
Respiratory Drugs		
Salmeterol	Salmeterol C _{max} (↑), AUC (↑↑↑↑) ^a	Not recommended during and for 2 weeks after treatment with itraconazole. Increased risk of salmeterol-related adverse reactions ^c .
SSRIs, Tricyclics and Related Antidepressants		
Reboxetine Venlafaxine	Reboxetine C _{max} (↔), AUC (↑) ^a Venlafaxine C _{max} (↑), AUC (↑) ^a	Use with caution, monitor for reboxetine/venlafaxine adverse reactions ^c , dose reduction of reboxetine/venlafaxine may be necessary.
Urologic Drugs		
Avanafil	Avanafil C _{max} (↑↑), AUC (↑↑↑↑) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk avanafil-related adverse reactions, such as priapism, visual problems and sudden loss of hearing.
Dapoxetine	Dapoxetine C _{max} (↑), AUC (↑) ^a	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk for dapoxetine-related adverse reactions, such as orthostatic hypotension and ocular effects.

Darifenacin Vardenafil	Darifenacin C _{max} (↑↑↑), AUC (↑↑↑ to ↑↑↑↑) ^a Vardenafil C _{max} (↑↑), AUC (↑↑↑↑) ^a	Not recommended during and for 2 weeks after treatment with itraconazole. Increased risk of darifenacin/vardenafil-related adverse reactions ^c .
Dutasteride Imidafenacin Oxybutynin Sildenafil (erectile dysfunction) Tadalafil (erectile dysfunction and benign prostatic hyperplasia) Tolterodine Udenafil	Dutasteride conc increase (extent unknown) ^{a,b} Imidafenacin C _{max} ↑, AUC ↑ Oxybutynin conc increase ↑ ^b Sildenafil C _{max} (↑↑), AUC (↑↑ to ↑↑↑↑) ^a Tadalafil C _{max} (↑), AUC (↑↑) ^a Tolterodine C _{max} (↑ to ↑↑), AUC (↑↑) ^a in poor metabolizers of CYP2D6 Udenafil C _{max} (↑), AUC (↑↑) ^a	Use with caution, monitor for urologic drug adverse reactions ^c , dose reduction of the urologic drug may be necessary; refer to the dutasteride label for specific actions to be taken. (For sildenafil and tadalafil, see also <i>Cardiovascular Drugs, Miscellaneous Drugs and other substances</i> .)
Fesoterodine	Fesoterodine C _{max} (↑↑), AUC (↑↑) ^a	Contraindicated in patients with moderate to severe renal or hepatic impairment, during and for 2 weeks after treatment with itraconazole. Increased risk of fesoterodine-related adverse reactions, such as severe anticholinergic effects. Use with caution in other patients: monitor for fesoterodine adverse reactions ^c , dose reduction of fesoterodine may be necessary.
Solifenacin	Solifenacin C _{max} (↑), AUC (↑↑) ^a	Contraindicated in patients with severe renal or moderate to severe hepatic impairment, during and for 2 weeks after treatment with itraconazole. Increased risk of solifenacin-related adverse reactions, such as anticholinergic effects and QT prolongation. Use with caution in other patients, monitor for solifenacin drug adverse reactions ^c , dose reduction of solifenacin may be necessary.

Miscellaneous Drugs and Other Substances		
Alitretinoin (oral) Cabergoline Cannabinoids Cinacalcet	Alitretinoin $C_{\max}$ ($\uparrow$), AUC ($\uparrow$) ^a Cabergoline $C_{\max}$ ($\uparrow\uparrow$), AUC ($\uparrow\uparrow$) ^a Cannabinoids conc increase, extent unknown but likely ($\uparrow\uparrow$) ^a Cinacalcet $C_{\max}$ ($\uparrow\uparrow$), AUC ($\uparrow\uparrow$) ^a	Use with caution, monitor for alitretinoin/cabergoline/cannabinoids/cinacalcet drug adverse reactions ^c , dose reduction of alitretinoin/cabergoline/cannabinoids/cinacalcet may be necessary.
Colchicine	Colchicine $C_{\max}$ ($\uparrow$), AUC ($\uparrow\uparrow$) ^a	Contraindicated in patients with renal or hepatic impairment, during and for 2 weeks after treatment with itraconazole. Increased risk of colchicine-related adverse reactions, such as decreased cardiac output, cardiac arrhythmias, respiratory distress and bone marrow depression. Not recommended in other patients, during and for 2 weeks after treatment with itraconazole. Increased risk of colchicine-related adverse reactions ^c .
Eliglustat	CYP2D6 EMs: Eliglustat $C_{\max}$ ($\uparrow\uparrow$), AUC ($\uparrow\uparrow$) ^a Higher increases are expected in CYP2D6 IMs/PMs and upon coadministration with a CYP2D6 inhibitor.	Contraindicated in CYP2D6 EMs taking a strong or moderate CYP2D6 inhibitor / CYP2D6 IMs and PMs, during and for 2 weeks after treatment with itraconazole. Increased risk of eliglustat-related adverse reactions such as prolongation of the PR, QTc, and/or QRS cardiac interval, and cardiac arrhythmias. Use with caution in CYP2D6 EMs, monitor for eliglustat adverse reactions ^c , dose reduction of eliglustat may be necessary.
Ergot alkaloids	Ergot alkaloids conc increase (extent unknown) ^{a,b}	Contraindicated during and for 2 weeks after treatment with itraconazole. Increased risk of ergot alkaloid-related adverse reactions, such as ergotism. (see also <i>Antimigraine Drugs</i>)
Galantamine	Galantamine $C_{\max}$ ($\uparrow$), AUC ($\uparrow$) ^a	Use with caution, monitor for galantamine adverse reactions ^c . Dose reduction of galantamine may be necessary.

Ivacaftor	Ivacaftor C _{max} (↑↑), AUC (↑↑↑) ^a	Use with caution, monitor for ivacaftor adverse reactions ^c , dose reduction of ivacaftor may be necessary.
Lumacaftor/Ivacaftor	Ivacaftor C _{max} (↑↑), AUC (↑↑) ^a Lumacaftor C _{max} (↔), AUC (↔) ^a Itraconazole conc decrease, extent unknown but likely ↓↓↓	Not recommended from 2 weeks before, during and for 2 weeks after treatment with itraconazole. Itraconazole efficacy may be reduced and increased risk of ivacaftor-related adverse reactions ^c .
Vasopressin Receptor Antagonists		
Conivaptan Tolvaptan	Conivaptan C _{max} (↑↑), AUC (↑↑↑) ^a Tolvaptan C _{max} (↑↑), AUC (↑↑↑) ^a	Not recommended during and for 2 weeks after treatment with itraconazole. Increased risk of conivaptan/tolvaptan-related adverse reactions ^c .
Mozavaptan	Mozavaptan C _{max} ↑, AUC ↑↑	Use with caution, monitor for mozavaptan adverse reactions ^c , dose reduction of mozavaptan may be necessary.

*CYP3A4 inhibitors (including itraconazole) may increase systemic contraceptive hormone concentrations.

EMs: extensive metabolizers; IMs: intermediate metabolizers, PMs: poor metabolizers; TdP: Torsade de Pointes

Note:

Average increase:

↑: <100% (i.e. <2-fold);

↑↑: 100-400% (i.e. ≥2-fold to <5-fold);

↑↑↑: 400-900% (i.e. ≥5-fold and <10-fold);

↑↑↑↑: ≥10-fold;

Average decrease:

↓: <40%;

↓↓: 40-80%;

↓↓↓: >80%;

No effect: ↔;

For the effect (middle column) the name of the parent drug is stated, even when the effect is related to the active moiety or the active metabolite of a prodrug.

^a For drugs with arrows between brackets, the assessment was based on the mechanism of interaction and clinical drug interaction information with ketoconazole or other strong CYP3A4 inhibitors and/or inhibitors of P-glycoprotein or BCRP, modelling techniques, case reports and/or *in vitro* data. For the other drugs listed, the assessment was based on clinical drug interaction information with itraconazole.

^b Pharmacokinetic parameters were not available.

^c Please consult the corresponding label for information on drug-related adverse reactions

Pediatric population

Interaction studies have only been performed in adults.

Incompatibilities

None known.

ADVERSE REACTIONS

Throughout this section, adverse reactions are presented. Adverse reactions are adverse events that were considered to be reasonably associated with the use of itraconazole based on the comprehensive assessment of the available adverse event information. A causal relationship with itraconazole cannot be reliably established in individual cases. Further, because clinical trials are conducted under widely varying conditions, adverse reaction rates observed in the clinical trials of a drug cannot be directly compared to rates in the clinical trials of another drug and may not reflect the rates observed in clinical practice.

Clinical trial data

The safety of Itraconazole Oral Solution was evaluated in 889 patients who participated in six double-blind and four open-label clinical trials. Of the 889 patients treated with Itraconazole Oral Solution, 624 patients were treated with Itraconazole Oral Solution during the double-blind trials. All 889 patients received at least one dose of Itraconazole Oral Solution for the treatment of oropharyngeal and esophageal candidiasis and provided safety data. Adverse reactions reported for $\geq 1\%$ of patients treated with Itraconazole Oral Solution in these clinical trials are shown in Table 1.

Table 1: Adverse Reactions Reported by $\geq 1\%$ of Patients Treated with Itraconazole Oral Solution in 10 Clinical Trials

System Organ Class Adverse Reaction	Itraconazole oral solution % (N=889)
Nervous System Disorders	
Headache	3.6
Dysgeusia	1.5
Dizziness	1.1
Respiratory, Thoracic and Mediastinal Disorders	
Cough	1.8
Gastrointestinal Disorders	
Diarrhea	9.1
Nausea	8.2
Vomiting	5.2
Abdominal pain	4.5
Dyspepsia	1.0

Skin and Subcutaneous Tissue Disorders	
Rash	2.5
General Disorders and Administration Site Conditions	
Pyrexia	5.2

Adverse reactions that occurred in <1% of patients treated with Itraconazole Oral Solution in these clinical trials are listed in Table 2.

Table 2: Adverse Reactions Reported by <1% of Patients Treated with Itraconazole Oral Solution in 10 Clinical Trials	
System Organ Class	Adverse Reaction
Blood and Lymphatic System Disorders	
	Leukopenia
	Thrombocytopenia
Immune System Disorders	
	Hypersensitivity
Metabolism and Nutrition Disorders	
	Hypokalemia
Nervous System Disorders	
	Hypoesthesia
	Neuropathy peripheral
	Paresthesia
Ear and Labyrinth Disorders	
	Tinnitus
Cardiac Disorders	
	Cardiac failure
Gastrointestinal Disorders	
	Constipation
Hepatobiliary Disorders	
	Hepatic failure
	Hyperbilirubinemia
Skin and Subcutaneous Tissue Disorders	
	Pruritus
	Urticaria
Musculoskeletal and Connective Tissue Disorders	
	Arthralgia
	Myalgia
Reproductive System and Breast Disorders	
	Menstrual disorder
General Disorders and Administration Site Conditions	
	Edema

The following is a list of additional adverse reactions associated with itraconazole that have been reported in clinical trials of Itraconazole capsules and/or Itraconazole injection, excluding the adverse reaction term “Injection site inflammation” which is specific to the injection route of administration.

Infections and Infestations: Sinusitis, Upper respiratory tract infection, Rhinitis

Blood and Lymphatic System Disorders: Granulocytopenia

Immune System Disorders: Anaphylactoid reaction

Metabolism and Nutrition Disorders: Hyperglycemia, Hyperkalemia, Hypomagnesemia

Psychiatric Disorders: Confusional state

Nervous System Disorders: Somnolence

Cardiac Disorders: Left ventricular failure, Tachycardia

Vascular Disorders: Hypertension, Hypotension

Respiratory, Thoracic and Mediastinal Disorders: Pulmonary edema, Dysphonia

Gastrointestinal Disorders: Gastrointestinal disorder, Flatulence

Hepatobiliary Disorders: Hepatitis, Jaundice, Hepatic function abnormal

Skin and Subcutaneous Tissue Disorders: Rash erythematous, Hyperhidrosis

Renal and Urinary Disorders: Renal impairment, Pollakiuria, Urinary incontinence

Reproductive System and Breast Disorders: Erectile dysfunction

General Disorders and Administration Site Conditions: Generalized edema, Face edema, Chest pain, Pain, Fatigue, Chills

Investigations: Alanine aminotransferase increased, Aspartate aminotransferase increased, Blood alkaline phosphatase increased, Blood lactate dehydrogenase increased, Blood urea increased, Gamma-glutamyltransferase increased, Hepatic enzyme increased, Urine analysis abnormal

Pediatrics

The safety of Itraconazole Oral Solution was evaluated in 250 pediatric patients aged 6 months to 14 years who participated in five open-label clinical trials. These patients received at least one dose of Itraconazole Oral Solution for prophylaxis of fungal infections or for treatment of oral thrush or systemic fungal infections and provided safety data.

Based on pooled safety data from these clinical trials, the very common reported adverse reactions in pediatric patients were Vomiting (36.0%), Pyrexia (30.8%), Diarrhea (28.4%), Mucosal inflammation (23.2%), Rash (22.8%), Abdominal pain (17.2%), Nausea (15.6%), Hypertension (14.0%), and Cough (11.2%). The nature of adverse reactions in pediatric patients is similar to that observed in adult subjects, but the incidence is higher in the pediatric patients.

Post-marketing data

In addition to the adverse reactions reported during clinical studies and listed above, the following adverse reactions have been reported during post-marketing experience (Table 3). In the table, the frequencies are provided according to the following convention:

Very common $\geq 1/10$

Common $\geq 1/100$ and $< 1/10$

Uncommon $\geq 1/1000$ and $< 1/100$

Rare $\geq 1/10000$ and $< 1/1000$

Very rare $< 1/10000$, including isolated reports.

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In Table 3, adverse reactions are presented by frequency category based on spontaneous reporting rates.

Table 3: Adverse Reactions Identified During Post-Marketing Experience with Itraconazole by Frequency Category Estimated from Spontaneous Reporting Rates

Immune System Disorders	
<i>Very rare</i>	Serum sickness, Angioneurotic edema, Anaphylactic reaction
Metabolism and Nutrition Disorders	
<i>Very rare</i>	Hypertriglyceridemia
Nervous System Disorders	
<i>Very rare</i>	Tremor
Eye Disorders	
<i>Very rare</i>	Visual disturbances (including diplopia and vision blurred)
Ear and Labyrinth Disorders	
<i>Very rare</i>	Transient or permanent hearing loss
Cardiac Disorders	
<i>Very rare</i>	Congestive heart failure
Respiratory, Thoracic and Mediastinal Disorders	
<i>Very rare</i>	Dyspnea
Gastrointestinal Disorders	
<i>Very rare</i>	Pancreatitis
Hepatobiliary Disorders	
<i>Very rare</i>	Serious hepatotoxicity (including some cases of fatal acute liver failure)
Skin and Subcutaneous Tissue Disorders	
<i>Very rare</i>	Toxic epidermal necrolysis, Stevens-Johnson syndrome, Acute generalized exanthematous pustulosis, Erythema multiforme, Exfoliative dermatitis, Leukocytoclastic vasculitis, Alopecia, Photosensitivity
Investigations	
<i>Very rare</i>	Blood creatine phosphokinase increased

OVERDOSE

Symptoms and signs

In general, adverse events reported with overdose have been consistent with those reported for itraconazole use. (See *Adverse Reactions*.)

Treatment

In the event of an overdose, supportive measures should be employed.

It is advisable to contact a poison control center to determine the latest recommendations for the management of an overdose.

Itraconazole cannot be removed by hemodialysis.

No specific antidote is available.

DOSAGE AND ADMINISTRATION

For optimal absorption, JAMP Itraconazole Oral Solution should be taken without food (patients are advised to refrain from eating for at least 1 hour after intake).

For the treatment of oral and/or esophageal candidosis, the oral solution should be swished around the oral cavity (approx. 20 seconds) and swallowed. There should be no rinsing after swallowing.

Treatment of oral candidosis

200 mg (2 measuring cups, i.e. 20 mL) per day in two intakes, or alternatively in one intake, for 1 week. If there is no response after 1 week, treatment should be continued for another week.

Treatment of esophageal candidosis

100 mg (1 measuring cup, i.e. 10 mL) daily for a minimum treatment of three weeks. Treatment should continue for 2 weeks following resolution of symptoms. Doses up to 200 mg (2 measuring cups, i.e. 20 mL) per day may be used based on the clinical response of the patient.

Treatment of fluconazole resistant oral and/or esophageal candidosis

100 to 200 mg (1 - 2 measuring cups) twice daily for 2 weeks. If there is no response after 2 weeks, treatment should be continued for another 2 weeks. The 400 mg daily dose should not be used for longer than 14 days if there are no signs of improvement.

Prophylaxis of fungal infections

5 mg/kg per day administered in two intakes. In clinical trials, prophylaxis treatment was started immediately prior to the cytostatic treatment and generally one week before transplant procedure. Treatment was continued until recovery of neutrophils (i.e., > 1000 cells/ μ L).

Storage

Store at or below 25°C.

Keep out of reach of children.

Shelf Life

Unopened bottle: 24 months

Opened bottle: Discard 28 days after opening.

Nature and Contents of Container

150 ml amber glass bottle, with child-resistant polypropylene screw cap having a LDPE internal coating.

Manufacturer

Kleva Pharmaceutical S.A
189, Parnithos Avenue
Acharnai Attiki
Greece, 13675

Product Registration Holder

Pharmaforte (M) Sdn. Bhd.
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